

1-Introduction

The objective of this project is to clean and prepare the dataset by identifying missing values, checking duplicates, and validating data formats.

2-Missing Values

A	B	C	D	E	F	G	H	I	J	K	L	M	N
OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2		Email	268.56
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3		Google	473.96
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
ORD200017	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3		Referral	423.4
ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.44
ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120
ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6		Google	1371.8
ORD200022	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093
ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.9
ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6		Referral	297.75
ORD200027	2024-02-22	C97593	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	472.74

missing values were identified in the CouponCode column and replaced with " 309 No Coupon".

A	B	C	D	E	F	G	H	I	J	K	L	M	N
OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FREESHIP	Facebook	747.75
ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	No Coupon	Email	268.56
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3	No Coupon	Google	473.96
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
ORD200017	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3	No Coupon	Referral	423.4
ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.44
ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120
ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	No Coupon	Google	1371.8
ORD200022	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FREESHIP	Referral	1093
ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.9
ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	No Coupon	Referral	297.75
ORD200027	2024-02-22	C97593	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	472.74

3-Duplicate Check

No duplicate records or duplicate Order IDs were found.

A	B	C	D	E	F	G	H	I	J	K	L	M	N
OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FRESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	SAVE10	Facebook	664.42
ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FRESHIP	Facebook	747.75
ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	No Coupon	Email	268.56
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FRESHIP	Referral	361
ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FRESHIP	Facebook	604.47
ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3	No Coupon	Google	473.96
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FRESHIP	Email	1067.62
ORD200017	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3	No Coupon	Referral	423.4
ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.44
ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120
ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6	No Coupon	Google	1371.8
ORD200022	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FRESHIP	Referral	1093
ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.9
ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6	No Coupon	Referral	297.75
ORD200027	2024-02-22	C97593	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	472.74

4. Date Validation

All date values were reviewed and found to be correctly formatted.

B1	A	B	C	D	E	F	G	H	I
	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderSt
1	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped
2	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped
3	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled
4	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned
5	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered
6	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped
7	ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned
8	ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped
9	ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled
0	ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped
1	ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned
2	ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned
3	ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending
4	ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned
5	ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending
6	ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered
7	ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending
8	ORD200017	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped
9	ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending
0	ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned
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2	ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered
3	ORD200022	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending
4	ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled
5	ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered
6	ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered
7	ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered
8	ORD200027	2024-02-22	C97593	Printer	1	472.74	881 Main St	Online	Shipped

6. Conclusion

The dataset was successfully cleaned and prepared for analysis. Missing values were handled, duplicate records were checked, and data formats were validated.